

CS 203: Software Tools & Techniques for AI
IIT Gandhinagar
Sem-II - 2024-25

Lab Assignment 9

Total: 10 Marks

Submission deadline: Sunday, 13/04/2025 11:59:59 PM

Submission guidelines:

1. Implement the code in the jupyter notebook and share it during submission.
2. Add all screenshots of the results in a pdf file and share.
3. The deadline will not be extended at any cost.

Note: Submitting this assignment solution confirms that you will follow the IITGN's honor code. We shall strictly penalize the submissions containing plagiarized text/code.

Objective

This assignment will introduce students to **CI/CD for Machine Learning** using [MLRun](#), a powerful orchestration and automation framework for deploying ML pipelines. Students will learn to create a function, deploy it using Kubernetes/Nuclio, and set up a simple retraining pipeline.

Dataset Preparation

Get the breast cancer dataset using “from sklearn.datasets import load_breast_cancer”.

Installation guide

Features to Implement

1. **Create a mlrun project:**
 - a. Use mlrun `new_project`. [10%]

2. Create the following Python script:

- a. Data_prep.py: This fetches data using sklearn.datasets import `load_breast_cancer`. (Add screenshot of the code)[10 %]
- b. Trainer.py: Split the data into train test (10% test data). Train a model using the training data. Use Random forest classifier. Wrap the model with `apply_mlrun` from `mlrun.frameworks.sklearn`. (Add screenshot of the code)[10 %]
- c. serving.py: Create a model class that will inherit from `mlrun.serving.V2ModelServer`, enabling automatic support for model lifecycle methods like `load()` and `predict()`. (Add screenshot of the code)[10 %]
- d. workflow.py: Create a Python script that defines an MLRun pipeline using the `@dsl.pipeline` decorator. It should include the following steps:
 - i. Data Ingestion(Add screenshot of the code) [3 %]
 - ii. Model Training: Experiment with different hyperparameters(`n_estimators`: [10, 100, 200], `max_depth`=[2, 5, 10]). Keep max accuracy as selector. (Add screenshot of the code)[4 %]
 - iii. Model Deployment: Deploy the model into the base kubernetes image(`mlrun/mlrun`) using `mlrun.deploy_function` (Add screenshot of the code)[3 %]

3. Run the workflow using `project.run`:

- a. Add the screenshot of the workflow graph [10 %]
- b. Add the screenshot of the Data_prep artifact and take the screenshot of the dataframe. [10 %]
- c. Add the screenshot of the confusion-matrix artifact of train.py [10 %]
- d. Add the screenshot of feature selection artifact. [10 %]

4. Code Quality:

- a. Maintain **clean**, **modular**, and readable code. [10 %]

*****Best of Luck*****